UNDER REVIEW

RESCIENCE C

Replication

[RE] A Cyber-Physical System for Freeway Ramp Meter Signal Control Using Deep Reinforcement Learning in a Connected Environment

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A replication of *"A Cyber-Physical System for Freeway Ramp Meter Signal Control Using Deep Reinforcement Learning in a Connected Environment"* from Hou et al. (2021), published in the *2021 IEEE Intelligent Transportation Systems Conference (ITSC)*, Indianapolis, USA, September 19–21, 2021

Replication Summary

Scope of Replication — The objective of this work is to reproduce the results of the reference study [1] and verify the claim that the reinforcement learning algorithms Ape-X DQN, PPO and A3C show superior performance for freeway ramp meter signal control in the metrics traffic throughput, vehicle speed, number of stops per vehicle, fuel efficiency per vehicle and per-vehicle emission rates for CO_2 and NO_x than the given baseline controllers ALINEA, no ramp meter and fixed-time controller in both mixed near-congested/congested traffic situations and extremely congested traffic situations. Furthermore, we aim to confirm that PPO shows better results in all of the aforementioned assessment criteria than the other reinforcement learning algorithms, while showing a convergence trend similar to Ape-X DQN and one that is approximately 15 times faster than A3C. To achieve this, we implemented a simulation environment in Python, together with the three baseline controllers (ALINEA, no ramp meter and fixed-time control) and the three RL-based controllers (Ape-X DQN, PPO and A3C), which we subsequently used to compare our obtained results with the original claims.

Methodology — Since the authors of the reproduced study did not provide any source code, the modelling of the environment, the route generation, the network, the reward function, the state computation, the three baseline controllers, the three reinforcement learning algorithms and also the training and evaluation scripts were newly implemented based on the paper description. For Ape-X DQN, PPO and A3C we made use of the reinforcement learning library RLlib [2] to remain consistent with the original study and the microscopic traffic simulations were conducted using the SUMO software (Simulation of Urban Mobility) together with TraCI (Traffic Control Interface) [3].

Results — TODO –> new results have been found, this section will be updated after final evaluation! Our replication findings do not match the original work from [1]. With their described parameters, the baseline controller (ALINEA, no ramp meter and fixed-time ramp meter) already showed significant deviations in the evaluation metrics. Furthermore, the hypothesis that the reinforcement learning policies outperform the baseline

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Code is available at https://github.com/DerKevinRiehl/eth_replication_project_rl_rm_work/tree/2021_Hou_et_al.

controllers could not be reproduced. Moreover, we were not able to provide evidence that PPO surpasses Ape-X DQN and A3C in our measurements. However, we can confirm that PPO converges at a similar rate as Ape-X DQN and that A3C takes about 15 times longer.

What was easy — The implementation of the three baseline-feedback controllers (ALINEA, no ramp meter and fixed-time controller) was straightforward, especially no ramp meter and fixed-time controller. The reason for this is that the paper provided both formulas and parameter values for ALINEA, while also specifying the cycle length for the fixed-time controller.

What was difficult — Implementing the environment with all its details required significant effort.

1 Introduction

With the increase of urbanization and global population growth, cities are expected to house an additional 2.5 billion people by 2050 [4]. This rising number of people in cities will inevitably result in higher traffic demand in metropolitan areas. Generally, travel demand surpasses road capacity on a regular basis, especially with poorly coordinated traffic signal phases and the resulting congestion negatively impacts not only a nation's economy, but also the individual's quality of life [5].
Freeway bottlenecks, particularly on-ramp merging areas, represent one of the major causes of congestion in urban transportation networks [5]. To address this issue, ramp metering strategies have been widely adopted to regulate the flow of vehicles entering freeways and maintain stable traffic conditions, where the red-to-green cycles are driven by a control algorithm [6, 7]. Classical approaches that have been introduced in literature include fixed-time controllers, feedback-based methods such as ALINEA [8], fuzzy-control [6, 7], neural-network-based approaches [9, 10, 11, 12], model-based predictive controllers [13] as well as techniques combining multiple control paradigms [14].
However, freeway traffic systems remain highly dynamic, stochastic and nonlinear, making it difficult for traditional control methods to generalize across varying traffic conditions. In recent years, reinforcement learning (RL) has demonstrated impressive performance in solving complex decision-making problems through direct interaction with an environment. These domains range from playing games [15, 16, 17] to robotics [18, 19] as well as fields like energy systems [20, 21] and intelligent infrastructure management [22, 23]. Encouraged by these successes, RL has increasingly been adopted in mobility applications such as traffic signal systems [24, 25, 26, 27, 28], decision making in autonomous vehicles [29, 30, 31, 32, 33, 34], dynamic speed regulation [35, 36] and adaptive road pricing strategies [37, 38].
Recently, several studies have also investigated the use of RL for freeway ramp metering control [39, 40]. For instance, Rezaee et al. [41] proposed a RL-based approach that outperformed the widely used ALINEA controller.
Building upon these developments, this work is a reproduction study of the paper by Hou et al. [1], in which three different RL algorithms are applied to the problem of freeway ramp metering control and compared to three baseline controllers.

2 Summary of Original Paper

The original work *"A Cyber-Physical System for Freeway Ramp Meter Signal Control Using Deep Reinforcement Learning in a Connected Environment"* by Hou et al. [1] investigates the application of deep reinforcement learning to freeway ramp metering control. The

study compares three conventional baseline controllers — no ramp metering, a fixed-time controller, and ALINEA — with three deep RL algorithms, namely Ape-X DQN [42], PPO [43], and A3C [44] on a freeway network consisting of a two-lane mainline and a single merging on-ramp.
The objective of the proposed controllers is to improve mobility and environmental efficiency under different traffic conditions. To evaluate controller performance, the study considers six different metrics that degrade under congestion.

- Operational Efficiency
  - Traffic Throughput (vehs/h)
  - Average Vehicle Speed (m/s)
  - Number of Stops per Vehicle
- Environmental Impact
  - Fuel Efficiency (mpg)
  - CO_2 Emissions (g/mi)
  - NO_x Emissions (mg/mi)

The paper reports that the average performance of the deep RL policies is superior to each of the baseline controllers across all aforementioned metrics in both mixed near-congested/congested and extremely congested traffic scenarios.

3 Implementation

3.1 Building the Simulation Environment

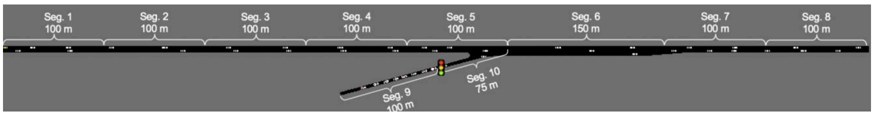

Figure 1. The simulated on-ramp merging area. Figure taken from [1].

Simulation Network — The simulation network consists of a two-lane freeway and a single-lane on-ramp merging into the mainline. The freeway is divided into eight segments (Seg. 1 - 8), each with a length of 100 m, except for the merging area (Seg. 6), which has an extended length of 150 m to enable smoother merging.
The on-ramp consists of additional segments, Seg. 9 and Seg. 10, with lengths of 100 m and 75 m, respectively. The ramp meter traffic signal, which is regulated by the implemented control algorithms, is located between Seg. 9 and Seg. 10. At Seg. 6, the network temporarily expands to three lanes as this is where the on-ramp lane gets merged to the freeway, before it tapers down to two lanes again.

3.2 Episode Logic and Traffic Flow Generation

To obtain statistically meaningful results, all controllers were evaluated over 20 simulation episodes (using the microscopic traffic simulator SUMO together with the TraCI interface [3]) and the reported metrics were averaged across all runs. Each episode consisted of 1000 simulation steps. Before the actual evaluation started, an additional warm-up phase of 40 simulation steps was introduced in order to establish realistic traffic conditions before measurements took place.
Two traffic scenarios were considered: mixed near-congested/congested traffic conditions and extremely congested traffic conditions. Following the assumptions made in the original study, the bottleneck capacity of the network using the default SUMO vehicle parameters was estimated to be between 3600 and 3700 veh/h.

In the mixed traffic scenario, the freeway inflow was sampled from a Gaussian distribution $\mathcal{N}(3000, 300)$ veh/h and then clipped to the interval $[0, 3900]$ veh/h. Similarly, the on-ramp inflow was generated independently from $\mathcal{N}(900, 300)$ veh/h and clipped to $[0, 1500]$ veh/h. In the extremely congested traffic scenario, the freeway inflow was fixed at 3900 veh/h, while the on-ramp inflow followed the same distribution as in the mixed traffic scenario, namely $\mathcal{N}(900, 300)$ veh/h clipped to the interval $[0, 1500]$ veh/h. To evaluate the controllers under varying traffic conditions, the traffic inflow on both the freeway and the on-ramp was updated every 100 simulation steps according to the above distributions.

Emission and Energy Model — For measuring the three metrics average vehicle fuel efficiency (mpg), CO_2 (g/mi) and NO_x (mg/mi), the default emission model in SUMO [45] was used, as no parameters related to the emission model were adjusted in the implementation.

3.3 Baseline Controllers

The following baseline controllers were implemented to benchmark the performance of the RL-based approaches.

1. **No Ramp Metering (No RM):** No control is applied, meaning the traffic signal will always be green throughout the entire simulation.

2. **Fixed-Time Control (Fixed):** Ramp metering is controlled by a fixed cycle consisting of a fixed red phase and a fixed green phase. In the original study the total cycle time is 4 seconds, with a 3-second red phase and a 1-second green phase.

3. **ALINEA:** Ramp metering is controlled by a variable cycle consisting of a variable red phase and a 1-second green phase. The cycle time c is recomputed every 15 seconds according to

$$c = \frac{N}{q_{\text{desire}} - q_{\text{freeway}}} \cdot 3600 \tag{1}$$

where:

N is the number of lanes on the ramp (in the given setup $N = 1$),
q_{desire} is the desired total traffic inflow at the merging area bottleneck (vph),
q_{freeway} is the traffic inflow on the freeway.

Moreover, the desired inflow at the bottleneck at time step t is given by

$$q_{\text{desire}}(t) = q_{\text{total}}(t-1) + C_f(n^* - n_{t-1}), \tag{2}$$

where:

$q_{\text{total}}(t-1)$ is the combined inflow of freeway and ramp at the previous time step $t-1$,
C_f is a sensitivity factor,
n^* is the critical number of vehicles at the merging area for maximum traffic outflow,
n_{t-1} is the number of vehicles at the merging area at time step $t-1$.

In the original work, the parameters n^* = 10 and C_f = 20 were used [1].

4 Reinforcement Learning

4.1 State Representation

The state is represented with a 40-dimensional vector with the following 3 parts:

1. The first 38 entries are derived from the mean speed and number of vehicles on each lane of each segment. As described earlier, the network is split into 10 different segments, hence the dimensionality for each network part is computed as:

$$n_{\text{segments}} \times n_{\text{lanes}} \times n_{\text{variables}},$$

   where $n_{\text{variables}} = 2$, corresponding to the mean speed and the number of vehicles:

$$\overbrace{7 \times 2 \times 2}^{(1)} + \overbrace{2 \times 1 \times 2}^{(2)} + \overbrace{1 \times 3 \times 2}^{(3)} = 38$$

   The first individual term (1) covers segments 1, 2, 3, 4, 5, 7 and 8. The second term (2) addresses the segments on the ramp, namely segments 9 and 10. The third term (3) covers the merging area with three lanes, namely segment 6.

2. The 39th entry is given by the ramp meter signal (red or green, so 0 or 1 respectively) of the previous simulation step.

3. The 40th and last entry of the state vector depicts the traffic throughput of the merging area bottleneck over the last 30 seconds.

4.2 Action Space

In the considered ramp metering scenario, the traffic signal regulates the vehicle flow from the on-ramp onto freeway with a two-state signal: red or green. Consequently, the action space is binary.

4.3 Reward Function

While congestion at the merging area (Seg. 6) and before the location of the ramp meter device (Seg. 9) should be heavily penalized, high vehicle speeds in the mentioned network parts should be rewarded, the reward function's definition is:

$$R(s_t, a_t) = \frac{\bar{v}_{6,9,10}}{s_{max}} + c_m + c_r \tag{3}$$

where:

- $\bar{v}_{6,9,10}$ is the mean speed on segments 6, 9, and 10.
- s_{max} denotes the maximum speed limit on the freeway.
- c_m is a penalty for congestion in the merging area (Seg. 6), defined as -1 if ≥ 5 vehicles have a speed < 1 m/s.
- c_r represents a penalty for ramp queueing (Seg. 9 and Seg. 10), set to -1 if ≥ 9 vehicles have a speed < 1 m/s on segment 9.

4.4 Building the RL Agents

To implement and train the agents, the RLlib library was used [2], leveraging 32 CPUs for parallel rollout workers. The neural network architecture remained consistent across all three algorithms, featuring a network with one hidden layer, consisting of 128 neurons and a non-linear activation function, namely tanh. The following parameters were specified in the original work:

Table 1. Hyperparameter settings for the RL algorithms

Parameter	Ape-X DQN	PPO	A3C
Sample size (per iteration)	1,000	32,000	500
Learning rate	5×10^{-4}	5×10^{-5}	1×10^{-4}
Exploration (ϵ)	$0.1 \rightarrow 10^{-4}$	–	–
Replay buffer size	5,000,000	–	–
Convergence (steps)	$\sim 20 \times 10^6$	$\sim 20 \times 10^6$	$\sim 300 \times 10^6$

Note: A dash (–) indicates that the parameter was not specified or is not applicable.

The original study claims that the reward curve of all three algorithms converged in training. More specifically, Ape-X DQN and PPO reward plateaued after about 20 000 000 training steps, whilst it took approximately 300 000 000 training steps for A3C.

5 Results

TODO

5.1 Critical Remarks

One major challenge in reproducing the work by [1] was identifying the right vehicle and controller parameters in order to match the reported traffic throughput and speed values more closely from the original paper, as the proposed default SUMO vehicle parameters did not yield results consistent with those described in their paper.

Additionally, for inflow values sampled from the Gaussian distributions, the generated values were clipped by setting the lower bound to 0. After extensive debugging, it was discovered that SUMO does not allow an inflow value of 0, which caused the episode to terminate immediately and consequently no result file to be written for that episode. Initially, this skewed the measurement results, as only 19 out of 20 episodes actually contributed to the evaluation metrics. This issue could have been considered by the original authors in order to ensure a more robust setup.

Moreover, in the network definition, the lengths of segments ending in a junction are automatically reduced by SUMO, as part of the segment is merged into the junction area. This behaviour was not taken into account by the paper, although additional details regarding whether and how this issue was handled could have been useful for the reproduction of their study.

Furthermore, the claim that only one car can pass the merging area per green phase in the fixed-time controller is not possible to implement, as the traffic signal state cannot be modified during the traci.simulationStep() function in SUMO. However, following the suggested cycle of 3 seconds red and 1 second green, it can be observed with the SUMO-GUI that not more than one car passes the traffic light during the green phase.

In addition, the training of PPO took significantly longer than reported in the original paper, requiring approximately twice the reported 5 wall-clock hours to reach 20 million training steps. This was observed even though all non-essential computations and logging within the environment were disabled to reduce the computational overhead and save resources, which was not further specified by the original authors. Although 32 CPUs were used for training, as stated in the paper, the difference in training times may stem from the fact that the exact hardware details used by Hou et al. were not disclosed.

Finally, the specified parameters in the original work for ALINEA and the cycle length for the fixed-time controller were not sufficient to accurately reproduce the results. With empirical experiments, better parameters were found for more accurate reproduction values.

6 Conclusion

TODO

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